Project Design Phase-II

Technology Stack(Architecture & Stack)

Date	4 March 2026
Team ID	SWUID202642221
Project Name	Gruha Alankara: Interior Design Platform with AI Design Agent and Augmented Reality Visualization
Maximum Marks	4 Marks

Technical Architecture

The system follows a **three-tier architecture** consisting of the **presentation layer, application layer, and data layer**. The **presentation layer** provides a web interface developed using **HTML, CSS, JavaScript, and React.js** where users upload room images and view design suggestions. The **application layer** uses **Python and FastAPI** to process requests and integrate AI services like **IBM Watsonx Vision** and **Groq API** for object detection and design recommendations. The **data layer** stores user data, images, and results using **MySQL with SQLAlchemy** and cloud storage. This architecture ensures the system is **scalable, efficient, and reliable** for AI-powered interior design assistance.

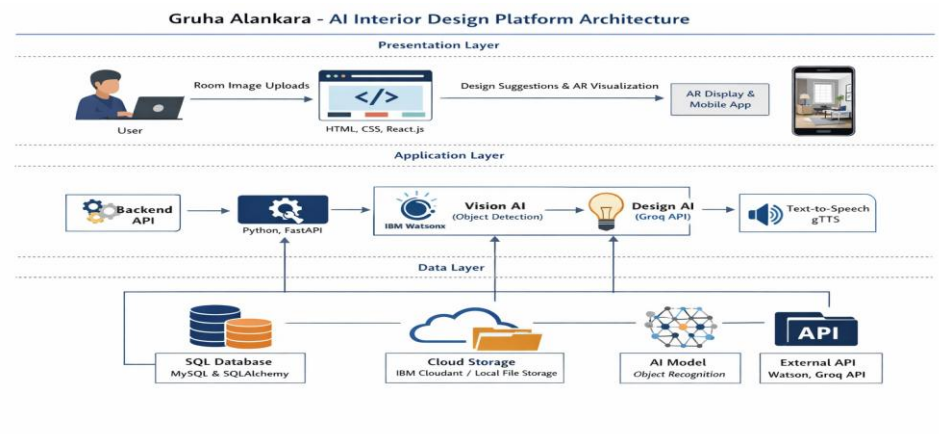

Table-1: System Components – Gruha Alankara AI Interior Design System

S.No	Component	Description	Technology
1	User Interface	How the user interacts with the application such as uploading room images and viewing AI design suggestions	HTML, CSS, JavaScript, React.js
2	Application Logic-1	Backend logic for processing user requests and managing system operations	Python, FastAPI
3	Application Logic-2	AI service used for speech processing and interaction	IBM Watson Speech-to-Text
4	Application	Conversational AI assistant to guide	IBM Watson

	Logic-3	users and answer queries	Assistant
5	Database	Stores user information, room images, and design suggestions	MySQL, SQLAlchemy
6	Cloud Database	Cloud-based database service for scalable data storage	IBM Cloudant / IBM DB2
7	File Storage	Stores uploaded images and generated design files	IBM Cloud Storage / Local File System
8	External API-1	API used for generating AI design suggestions	Groq API
9	External API-2	Vision AI API used for detecting objects in room images	IBM Watsonx Vision API
10	Machine Learning Model	AI model used for object detection and interior design analysis	Object Recognition Model
11	Infrastructure (Server / Cloud)	Deployment of application on local system or cloud server	Local Server / Cloud Foundry / Kubernetes

Table-2: Application Characteristics – Gruha Alankara AI Interior Design System

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	List of open-source frameworks used in the system	React.js, FastAPI, SQLAlchemy
2	Security Implementations	Security measures implemented for authentication and data protection	SHA-256 Encryption, HTTPS, IAM Controls
3	Scalable Architecture	System architecture designed to support increasing number of users	Three-Tier Architecture, Cloud Deployment
4	Availability	Ensures the system is accessible with minimal downtime	Cloud Hosting, Load Balancer
5	Performance	Design considerations to improve system response and processing speed	API Optimization, Caching

References

1. C4 Model for Software Architecture – Available at: https://c4model.com/
2. IBM Developer – *Online Order Processing System During Pandemic Architecture Pattern*. Available at: https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/
3. IBM – *Cloud Architecture Center*. Available at: https://www.ibm.com/cloud/architecture
4. Amazon Web Services – *AWS Architecture Center*. Available at: https://aws.amazon.com/architecture
5. Medium – *How to Draw Useful Technical Architecture Diagrams*. Available at: https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d